Mental Health Analysis: BRFSS 2024

Project Proposal Talking Points

1. What is This Project About?

What to say:

> "My project analyzes mental health outcomes across the US population using the CDC's Behavioral Risk Factor Surveillance System (BRFSS) 2024 dataset one of the largest public health surveys in the world with **449,514 respondents** and **301 variables**. My outcome variable is **MENTHLTH** the number of poor mental health days in the past 30 days and I examine how it varies across sex, age, income, and smoking status."

2. Descriptive Statistics — What Does the Data Look Like?

Your results:

- Mean = **5.09 days**, Median = **0 days**, SD = **8.98**

What to say:

> "The data is heavily right-skewed most Americans report zero poor mental health days, but a smaller group reports a very high number. This is why the mean (5.09) is much higher than the median (0). This skewness is actually an important finding in itself it tells us that mental health burden is concentrated in a subset of the population."

3. t-Test: Mental Health by Sex

Your results:

- Females: **5.72 days** vs Males: **4.36 days**
- t-statistic = **37.38**, p-value = **0** (essentially zero)
- 95% CI for difference: **(1.29, 1.43)**

What to say:

"I ran a Welch two-sample t-test to compare mental health days between males and females. The result is highly significant females report on average **1.36 more poor mental health days** per month than males. The p-value is effectively zero, meaning this difference is not due to chance. The 95% confidence interval of 1.29 to 1.43 days tells us we are very confident the true difference in the population lies in this range."

4. ANOVA: Mental Health by Income Level

Your results:

- Lowest income (<$15k): **8.40 days**
- Highest income (>$50k): **4.03 days**
- F-statistic = **1305.3**, p-value = **0**
- All pairwise income comparisons are significant (Tukey HSD)

What to say:

"A one-way ANOVA showed income level is a highly significant predictor of mental health. People earning less than $15k report **more than twice** the poor mental health days compared to those earning over $50k — 8.40 vs 4.03 days. The Tukey post-hoc test confirmed that **every single income group is significantly different from every other group**, meaning the effect of income on mental health is consistent and gradual — the more you earn, the fewer poor mental health days you report."

5. ANOVA: Mental Health by Smoking Status

Your results:

- Everyday smokers: **8.23 days**
- Never smokers: **4.36 days**
- F-statistic = **1556.9**, p-value = **0**

What to say:

"Smoking status is the strongest ANOVA predictor in my model. Everyday smokers report nearly **twice as many** poor mental health days as never-smokers — 8.23 vs 4.36 days. This is a statistically significant finding with a very large F-statistic. It's important to note this is a correlation, not causation smokers may use cigarettes as a coping mechanism for mental distress, or poor mental health may lead to higher smoking rates."

6. ANOVA: Mental Health by Age Group

Your results:

- 18–24 year olds: ~**7 days**
- 80+ year olds: ~**2.5 days**
- F-statistic = **868.2**, p-value = **0**

What to say:

> "Contrary to what many might expect, **younger adults report significantly more poor mental health days than older adults**. The trend is very clear mental health days decline consistently with age. The 18-24 group reports about 7 days per month while the 80+ group reports only 2.5 days. This is a fascinating public health finding that aligns with recent national trends showing a mental health crisis among Gen Z and younger millennials."

7. Correlation Analysis (Spearman)

Your results:

- AGE vs MENTHLTH: **-0.275** (strongest relationship)
- SEX vs MENTHLTH: **-0.106**
- INCOME vs MENTHLTH: **-0.097**
- SMOKER vs MENTHLTH: **-0.082**

What to say:

> "I used Spearman correlation because MENTHLTH is right-skewed. All four predictors are negatively correlated with mental health days meaning older age, higher income, and non-smoker status are all associated with fewer poor mental health days. **Age has the strongest correlation** at -0.275, followed by sex. The correlations are modest but meaningful given the complexity of mental health."

8. Multiple Linear Regression

Your results:

Predictor	Coefficient	Meaning
Sex (Female=1)	**-1.56**	Females report 1.56 more days

Age Group	**-0.48**	Each older age group = 0.48 fewer days
Income Level	**-0.79**	Each higher income bracket = 0.79 fewer days
Smoking Status	**-1.27**	Each step toward non-smoker = 1.27 fewer days
R-squared	**0.0797**	Model explains ~8% of variance

What to say:

> "My multiple regression model includes all four predictors simultaneously. All four are statistically significant with p-values essentially equal to zero. The model tells us:
>
> - **Being female adds 1.56 poor mental health days** per month even after controlling for age, income, and smoking.
> - **Each older age group reduces mental health days by 0.48** — confirming the age trend.
> - **Each higher income bracket reduces mental health days by 0.79** — confirming the income gradient.
> - The R-squared of 0.08 means these four variables explain about 8% of the variance. This is actually reasonable for mental health data since it's influenced by hundreds of factors beyond what we measured."

9. Key Limitations to Mention

What to say:

> "I want to be transparent about limitations:
>
> 1. **Causation vs Correlation** — this is observational data, so we cannot say income *causes* better mental health.
> 2. **Right-skewed outcome** — MENTHLTH is heavily skewed toward zero, which technically violates linear regression assumptions. A log transformation or Poisson regression could be explored.
> 3. **Self-reported data** — mental health days are self-reported, which introduces recall and social desirability bias.
> 4. **Missing data** — some variables had significant missingness, particularly EMPLOY1 (34%) and LSATISFY (56% in the full dataset)."

10. One-Sentence Project Summary

> "This project uses the BRFSS 2024 dataset to demonstrate that poor mental health in the US is significantly higher among **females, younger adults, lower-income groups, and smokers**, using a comprehensive statistical framework including hypothesis testing, confidence intervals, correlation analysis, and regression modeling."

Dataset: CDC BRFSS 2024 | n = 449,514 | Outcome: MENTHLTH (Poor Mental Health Days)

ANOVA — Analysis of Variance

ANOVA is a statistical test used to **compare the means of 3 or more groups** to see if at least one group is significantly different from the others.

Why Not Just Use Multiple t-Tests?

If you have 4 groups (like income levels), you'd need **6 separate t-tests** to compare all pairs. This increases the chance of a false positive. ANOVA tests all groups **at once** in a single test.

Simple Real-World Analogy

Imagine you want to know if **different diets affect weight loss**:

- Group A: Keto diet
- Group B: Vegan diet
- Group C: Mediterranean diet
- Group D: No diet

ANOVA answers: **"Is there ANY difference in weight loss across these 4 groups?"**

How It Works — In Plain English

ANOVA compares two things:

	Meaning
Variance BETWEEN groups	How different are the group means from each other?
Variance WITHIN groups	How spread out are values inside each group?

The **F-statistic** = Between-group variance ÷ Within-group variance

- **Large F** → groups are very different from each other → significant result
- **Small F** → groups are similar → not significant

In our Project

I used ANOVA 3 times:

Income vs Mental Health Days:

F = 1305.3, p ≈ 0

> The 5 income groups have very different mental health days; the result is highly significant.

Smoking vs Mental Health Days:

F = 1556.9, p ≈ 0

> The 4 smoking groups are significantly different everyday smokers suffer the most.

Age vs Mental Health Days:

F = 868.2, p ≈ 0

> The 13 age groups are significantly different; younger people suffer the most.

What ANOVA Doesn't Tell You

ANOVA only says **"at least one group is different"** — it does NOT say **which groups** differ. That's why you used the **Tukey HSD post-hoc test** after ANOVA — it identifies exactly which pairs of income groups are significantly different from each other.

> *"ANOVA tests whether the means of multiple groups are statistically equal, by comparing the variation between groups to the variation within groups using an F-statistic."*

Complete Statistics Guide

Using Your BRFSS 2024 Project as Examples

1. t-Test

What is it?

A t-test compares the **means of exactly 2 groups** to determine if they are significantly different from each other.

Simple Analogy

You want to know if **men and women sleep different amounts**. You measure 100 men and 100 women. A t-test tells you: *"Is this difference real, or just due to random chance?"*

Types of t-Test

Type	When to Use
One-sample	Compare one group to a known value
Two-sample (Independent)	Compare two separate groups
Paired	Compare same group before vs after

In Your Project

You used a **Welch Two-Sample t-Test** to compare mental health days between males and females:

Female mean = 5.72 days

Male mean = 4.36 days

Difference = 1.36 days

t-statistic = 37.38

p-value = ~0

The difference of 1.36 days is **statistically significant** — it is not due to chance.

One-Line Definition

> *"A t-test determines whether the means of two groups are statistically different by measuring how many standard errors apart they are."*

2. p-value & Hypothesis Testing

What is Hypothesis Testing?

Hypothesis testing is a formal framework for making decisions using data. It always involves two competing statements:

Hypothesis	Meaning
H0 (Null Hypothesis)	"Nothing is happening" / "No difference exists"
H1 (Alternative Hypothesis)	"Something is happening" / "A difference exists"

What is a p-value?

The p-value is the **probability of seeing your results (or more extreme) if H0 were true**.

Simple Analogy

You flip a coin 10 times and get 9 heads. You suspect the coin is rigged.

- H0: The coin is fair
- H1: The coin is rigged
- p-value: "If the coin were fair, how likely is it to get 9+ heads?" → Very unlikely → small p-value → reject H0

Decision Rule

p-value < 0.05 → Reject H0 → Result IS significant

p-value > 0.05 → Fail to reject H0 → Result is NOT significant

The threshold 0.05 means we accept a 5% chance of being wrong.

In Your Project

Every single test had p-value ≈ 0:

t-Test (Sex): p ≈ 0 → Females DO have more poor mental health days

ANOVA (Income): p ≈ 0 → Income groups ARE significantly different

ANOVA (Smoking): p ≈ 0 → Smoking groups ARE significantly different

ANOVA (Age): p ≈ 0 → Age groups ARE significantly different

Regression (Model): p ≈ 0 → All 4 predictors ARE significant

Common Misconception

> A p-value does NOT tell you the size of the effect — only whether it exists. With 449,514 people, even tiny differences become significant. That's why you also need to report **effect sizes and confidence intervals**.

One-Line Definition

> *"The p-value is the probability of observing your data if the null hypothesis were true — a small p-value means your results are unlikely to be due to chance."*

3. Confidence Intervals (CI)

What is it?

A confidence interval gives a **range of plausible values** for a population parameter (like a mean), based on your sample data.

Simple Analogy

A weather forecast says: *"Tomorrow's temperature will be between 72°F and 78°F."* That range is like a confidence interval — you're not sure of the exact value, but you're confident it falls within that range.

How to Interpret a 95% CI

> "If we repeated this study 100 times, 95 of those studies would produce a confidence interval containing the true population value."

Formula

CI = Mean ± (1.96 × Standard Error)

Lower bound = Mean - 1.96 × SE

Upper bound = Mean + 1.96 × SE

In Your Project

Overall MENTHLTH: Mean = 5.095 → 95% CI: (5.059, 5.131)

Female MENTHLTH: Mean = 5.720 → 95% CI: (5.669, 5.770)

Male MENTHLTH: Mean = 4.362 → 95% CI: (4.311, 4.412)

The female CI (5.669–5.770) and male CI (4.311–4.412) **do not overlap** — this visually confirms the difference is real.

Key Rule

> If two confidence intervals **do not overlap** → the groups are significantly different.
> If they **do overlap** → the difference may not be significant.

One-Line Definition

> *"A 95% confidence interval is a range we are 95% confident contains the true population mean."*

4. Spearman Correlation

What is it?

Correlation measures the **strength and direction of the relationship** between two variables. Spearman correlation is a version that works on **ranked data** — it doesn't assume the data is normally distributed.

Simple Analogy

Does studying more lead to better grades? If yes — there's a **positive correlation**. Does more stress lead to less sleep? If yes — there's a **negative correlation**.

Correlation Values

+1.0 = Perfect positive relationship

+0.5 = Moderate positive relationship

0.0 = No relationship

-0.5 = Moderate negative relationship

-1.0 = Perfect negative relationship

Why Spearman Instead of Pearson?

Pearson	Spearman
Assumes normal distribution	No distribution assumption
Works on raw values	Works on ranks
Best for continuous data	Best for skewed or ordinal data

Since MENTHLTH is heavily right-skewed (most values = 0), **Spearman was the right choice**.

In Your Project

AGE vs MENTHLTH: -0.275 → Strongest: older = fewer bad days

SEX vs MENTHLTH: -0.106 → Females tend to have more bad days

INCOME vs MENTHLTH: -0.097 → Higher income = fewer bad days

SMOKER vs MENTHLTH: -0.082 → Non-smokers have fewer bad days

All correlations are **negative and significant** — every predictor moves in the expected direction.

One-Line Definition

> *"Spearman correlation measures the monotonic relationship between two variables using their ranks, ranging from -1 (perfect negative) to +1 (perfect positive)."*

5. Linear Regression & R-squared

What is Linear Regression?

Linear regression finds the **best-fit line** through your data that predicts an outcome (Y) from one or more predictors (X).

Simple Analogy

You want to predict a person's **salary** based on their **years of experience**. Linear regression draws the best line through your data points and gives you:

Salary = 30,000 + (5,000 × Years of Experience)

For every 1 extra year → salary goes up $5,000.

Key Outputs

Output	Meaning
Intercept	Predicted Y when all X = 0
Coefficient	How much Y changes for 1 unit increase in X
p-value	Is this predictor significant?
R-squared	How much variance does the model explain?

What is R-squared?

R-squared (R^2) measures **how well your model fits the data** — the percentage of variance in Y explained by your predictors.

$R^2 = 0.00$ → Model explains nothing

$R^2 = 0.50$ → Model explains 50% of variance

$R^2 = 1.00$ → Model explains everything (perfect fit)

In Your Project

Model: MENTHLTH ~ SEX + AGE + INCOME + SMOKER

Intercept: 16.85 (baseline mental health days)

Sex: -1.56 (females report 1.56 MORE days)

Age: -0.48 (each older group = 0.48 FEWER days)

Income: -0.79 (each higher income = 0.79 FEWER days)

Smoker: -1.27 (each step toward non-smoker = 1.27 FEWER days)

$R^2 = 0.0797$ → Model explains ~8% of variance

Why is 8% R-squared OK?

Mental health is influenced by **hundreds of factors** — relationships, trauma, genetics, sleep, etc. Explaining 8% with just 4 demographic variables is actually meaningful in public health research.

One-Line Definition

> *"Linear regression models the relationship between an outcome and one or more predictors by fitting the best line through the data, with R-squared measuring how much variance the model explains."*

6. F-statistic

What is it?

The F-statistic is the test statistic used in ANOVA. It measures how much the **group means vary relative to how much individuals vary within groups**.

Formula (Conceptually)

$$F = \frac{\text{Variance BETWEEN groups}}{\text{Variance WITHIN groups}}$$

Simple Analogy

Imagine 3 classrooms taking a test:

- If all 3 classrooms score **very differently** from each other (high between-group variance) but students within each class score **similarly** (low within-group variance) → **Large F** → teaching method matters!
- If all classrooms score **about the same** → **Small F** → teaching method doesn't matter.

Interpreting F

Large F + small p-value → Groups ARE significantly different → Reject H0

Small F + large p-value → Groups are NOT significantly different → Fail to reject H0

In Your Project

Income ANOVA: F = 1305.3 → Enormous → income groups are very different

Smoking ANOVA: F = 1556.9 → Largest F → smoking has the biggest group differences

Age ANOVA: F = 868.2 → Very large → age groups are very different

All three F-statistics are massive, which is expected with 240,000+ observations.

One-Line Definition

> *"The F-statistic compares the variance between group means to the variance within groups — a large F value indicates the groups have meaningfully different means."*

7. Tukey HSD Post-hoc Test

What is it?

After ANOVA tells you *"at least one group is different"*, Tukey HSD (Honestly Significant Difference) tells you **exactly WHICH pairs of groups differ**.

Why is it Needed?

ANOVA is like a smoke alarm — it tells you there's a fire somewhere in the building. Tukey HSD is the firefighter who finds exactly which room is on fire.

Simple Analogy

ANOVA says: *"At least one income group has different mental health days."*
Tukey HSD says: *"<$15k vs >$50k differ. <$15k vs $35-50k differ. $15-25k vs >$50k differ..."* and so on for every pair.

How it Controls for Multiple Comparisons

Testing many pairs increases the chance of a false positive. Tukey HSD **adjusts the p-values** to account for this — it's more conservative than running individual t-tests.

In Your Project

All 10 income pair comparisons were significant (adj. p-value ≈ 0):

<$15k vs >$50k: difference = -4.37 days (biggest gap)

<$15k vs $35-50k: difference = -3.59 days

<$15k vs $25-35k: difference = -2.91 days

<$15k vs $15-25k: difference = -1.75 days (smallest gap)

...

Every income group is significantly different from every other — confirming a **perfect income gradient** in mental health.

One-Line Definition

> *"Tukey HSD is a post-hoc test that identifies which specific group pairs are significantly different after ANOVA, while controlling for multiple comparisons."*

8. Standard Deviation & Standard Error

Standard Deviation (SD)

What is it? Standard deviation measures **how spread out** individual values are around the mean.

Simple Analogy: Two pizza restaurants both have a mean delivery time of 30 minutes.

- Restaurant A: SD = 2 min → deliveries are always close to 30 min (consistent)
- Restaurant B: SD = 15 min → deliveries range from 15 to 45 min (inconsistent)

In Your Project:

MENTHLTH Mean = 5.09, SD = 8.98

The SD of 8.98 is larger than the mean of 5.09 — this confirms the data is highly spread out and right-skewed (many zeros but some very high values).

By Group:

<$15k income: Mean = 8.40, SD = 11.28 (very spread out)

>$50k income: Mean = 4.03, SD = 7.77 (less spread out)

Lower income groups show more variability in mental health — some do fine, but the worst cases are much worse.

Standard Error (SE)

What is it? Standard error measures **how precisely your sample mean estimates the true population mean**. It gets smaller as your sample size grows.

Formula:

$SE = SD \div \sqrt{n}$

Simple Analogy: If you ask 10 people their height, your average might be off. If you ask 10,000 people, your average will be much closer to the true population average. SE captures this — bigger sample = smaller SE = more precise estimate.

Difference Between SD and SE:

	Standard Deviation (SD)	Standard Error (SE)
Measures	Spread of individual values	Precision of the mean estimate
Affected by sample size	No	Yes — larger n = smaller SE
Used for	Describing data	Calculating confidence intervals

In Your Project: With n = 240,935 observations, your SE values are tiny:

$SE\ (Overall) = 8.98 \div \sqrt{240935} = 0.018$

This is why your confidence intervals are very narrow — your enormous sample gives very precise estimates.

One-Line Definitions

> *"Standard deviation measures how spread out individual data points are around the mean."*
> *"Standard error measures how precisely your sample mean estimates the true population mean it shrinks as sample size grows."*

Quick Reference Card

Concept	Purpose	Your Result

t-Test	Compare 2 group means	Female 5.72 vs Male 4.36 days ✓
p-value	Is result due to chance?	All p ≈ 0 → all significant ✓
Confidence Interval	Range for true mean	Overall CI: (5.059, 5.131) ✓
Spearman Correlation	Strength of relationship	Age strongest at -0.275 ✓
Linear Regression	Predict outcome from predictors	$R^2 = 0.08$ ✓
R-squared	How well model fits	8% variance explained ✓
F-statistic	ANOVA test statistic	Income F=1305, Smoking F=1557 ✓
Tukey HSD	Which group pairs differ?	All 10 income pairs significant ✓
Standard Deviation	Spread of individual values	SD = 8.98 (highly spread) ✓
Standard Error	Precision of mean estimate	SE = 0.018 (very precise) ✓

All examples drawn from BRFSS 2024 Mental Health Analysis | n = 449,514